The Courtois NeuroMod (CNeuroMod) Dataset

The CNeuroMod team

Funding: [List all funding sources with grant numbers.]

Abstract
[Add abstract here]

Introduction

Brain encoding models — trained to predict neural activity from the representations of artificial neural networks (ANNs) — have emerged as a powerful framework to study how the brain processes information [MISSING REF: e.g. Yamins & DiCarlo 2016, Schrimpf et al.]. Aligning artificial and biological representations through brain-augmented learning also shows early promise as a path toward more robust and generalizable AI, with models fine-tuned on brain activity demonstrating improved downstream task performance and faster learning from limited data (Bilgin et al., 2025).

The brain, however, is not a collection of independent modules. It is a multimodal, active system that continuously integrates perception, memory, language, and action. Building encoding models that capture this integrative capacity requires training data that spans a broad range of cognitive states — not just a single domain.

Current deep neuroimaging resources have made major strides in individual coverage, but each tends to optimize for depth within one cognitive domain — vision, language, or audition. Multi-participant datasets, on the other hand, prioritize population breadth at the cost of the individual depth needed to model within-subject brain organization. A resource combining deep individual sampling with genuinely multidomain cognitive coverage has yet to be assembled (Figure 1).

Several large neuroimaging initiatives have been designed for breadth — collecting data from thousands of subjects to enable population neuroscience and to train brain foundation models. The Human Connectome Project (HCP), UK Biobank, and OmniMouse exemplify this approach, offering wide coverage at the cost of shallow per-subject sampling.

A parallel tradition has pursued depth: recording from a small number of individuals for many hours under rich, naturalistic conditions. Figure 1 summarizes per-subject data volume across the most prominent resources in this space, spanning five cognitive categories — natural static image viewing, naturalistic video/audio/speech/reading, videogame play, controlled paradigms, and resting state — as well as multiple recording modalities (fMRI, EEG, MEG, iEEG) and physiological signals.

CNeuroMod stands out as the largest per-subject resource in naturalistic video and videogame categories by a wide margin. In the static image domain — dominated by the Natural Scenes Dataset (NSD), which exposes subjects to ~10,000 unique images — CNeuroMod offers ~4,300 unique image presentations per subject, placing it in the same tier while adding cross-domain coverage that NSD does not provide. Across physiology and brain recording modalities, CNeuroMod is consistently among the most richly instrumented resources.

The Courtois NeuroMod (CNeuroMod) project was designed to fill this gap. Over five years, six individuals were scanned for approximately 200 hours each (Boyle and Pinsard, n.d.), across a rich collection of 29 datasets spanning vision, language, memory, emotion, audition, and videogame play — the latter enabled by a custom fiber-optic controller developed specifically for the project. The dataset was assembled by a highly interdisciplinary team including specialists across all of these cognitive domains, and represents the largest and most cognitively diverse individual neuroimaging resource to date (Boyle and Pinsard, n.d.).

This paper describes the full CNeuroMod dataset. We present the design and rationale of the 29 datasets, provide evidence of high data quality across participants and modalities, and point to dedicated companion publications examining quality within each specific facet of the collection. We close with an overview of the many uses this dataset enables, from brain encoding and decoding to brain-augmented learning, multidomain cognitive modeling, and beyond.

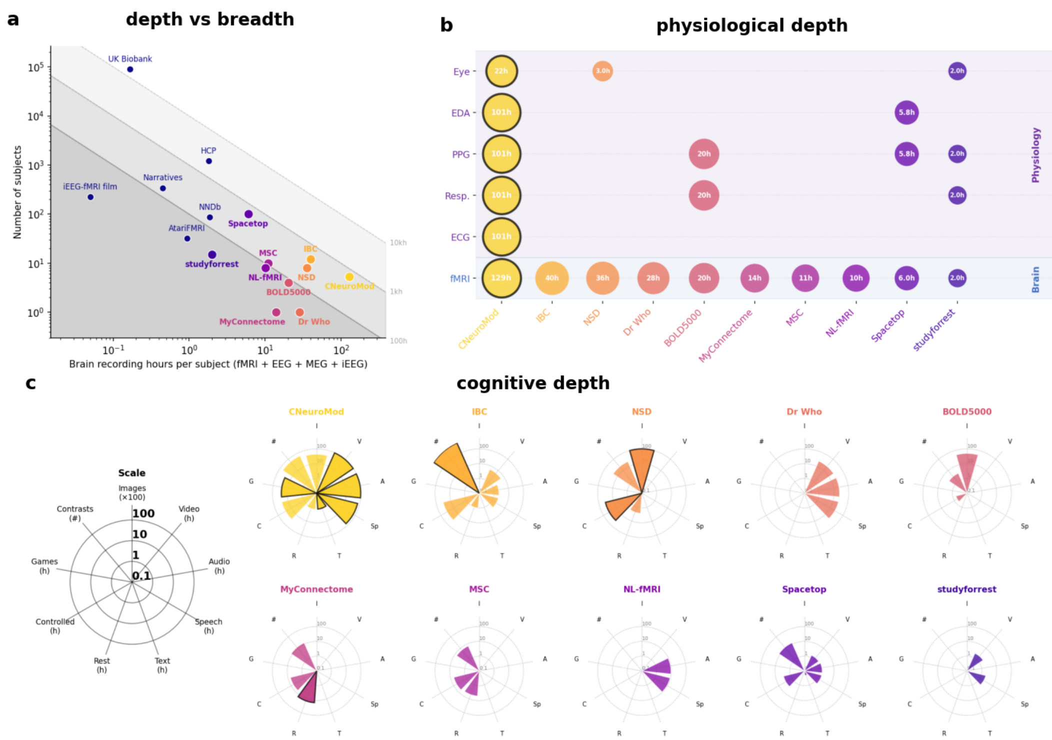

Figure 1. The dense NeuroAI dataset landscape. (a) Depth vs. breadth scatter plot: each dot is a neuroimaging dataset positioned by brain recording hours per subject (x-axis; fMRI, EEG, MEG, iEEG, and calcium imaging combined) and number of subjects (y-axis). Diagonal lines mark iso-total-hour contours. CNeuroMod (red) has the highest per-subject recording time of any public dataset. **(b)** Per-subject data volume for the largest dense NeuroAI datasets across brain recording modalities, task types, and physiological signals; bubble area scales logarithmically with data volume; black outline marks the largest resource in each column. **(c)** Cognitive depth radar charts for 11 dense NeuroAI datasets, showing per-subject coverage across cognitive domains. [MISSING REF: citations for all comparison datasets]

Data Overview

[Provide a high-level summary of what the dataset contains: number of sessions per participant, total scan time, tasks covered, and modalities available. May include summary tables and figures.]

Summary Statistics

[Tables summarizing sessions, hours of data, paradigm coverage per participant.]

Dataset Coverage

[Figures or descriptions illustrating the scope and completeness of the dataset.]

Data Acquisition

Participants

Six healthy adults (3 women, 2 men, and 1 trans woman; ages 31–47 at recruitment in 2018) consented to participate in the Courtois NeuroMod project for a minimum of five years. All participants were right-handed and reported good general health. Three were native francophones, one a native anglophone, and two were bilingual (English and French). Participants were excluded if they had visual or auditory impairments that would prevent perceiving stimuli in the scanner, major psychiatric or neurological conditions, or did not meet standard MRI compatibility criteria; all were required to have advanced English comprehension. Specific demographics for each participant are provided in [DATA RECORD TABLE]. Participants were scanned one to two times per week, except during personal holidays, illness, or periods of scanner unavailability. All procedures were approved by the ethics board of the CIUSSS du Centre-Sud-de-l'île-de-Montréal, and all participants provided written informed consent.

MRI Acquisition

Scanner and setup

All MRI data were collected at the Unité de Neuroimagerie Fonctionnelle (UNF), located at the Centre de Recherche de l'Institut Universitaire de Gériatrie de Montréal (CRIUGM), affiliated with the Université de Montréal. The scanner is a Siemens Prisma Fit (3T), equipped with a 2-channel transmit body coil and a 64-channel receive head/neck coil.

To minimize head motion, each participant wore a custom-fitted polystyrene foam headcase manufactured by Caseforge, milled from a 3D surface scan of the participant's head and shaped to fit the 64-channel coil.

Functional MRI

Functional MRI data were acquired using an accelerated simultaneous multi-slice gradient echo-planar imaging (EPI) sequence (Xu et al., 2013) developed at the Center for Magnetic Resonance Research (CMRR), University of Minnesota, as part of the Human Connectome Project (Glasser et al., 2016). The sequence was obtained through a concept-to-production (C2P) agreement and run with the following parameters: slice acceleration factor = 4, TR = 1.49 s, TE = 37 ms, flip angle = 52°, voxel size = $2 \times 2 \times 2$ mm, 60 slices, acquisition matrix 96×96. At the start of each session, a short acquisition (3 volumes) with reversed phase-encoding direction was collected to enable retrospective correction of B0 field inhomogeneity-induced distortions.

Brain anatomical MRI

Dedicated anatomical sessions were conducted approximately four times per year. Each session began with a 21 s localizer scan, followed by:

- T1-weighted MPRAGE 3D sagittal (6:38 min; TR = 2.4 s, TE = 2.2 ms, flip angle = 8°, 0.8 mm isotropic, R = 2)
- T2-weighted FSE (SPACE) 3D sagittal (5:57 min; TR = 3.2 s, TE = 563 ms, 0.8 mm isotropic, R = 2)
- Diffusion-weighted 2D axial (4:04 min; TR = 2.3 s, TE = 82 ms, flip angle = 78°, 57 slices, 2 mm isotropic, SMS = 3, b-max = 3000 s/mm^2; repeated with reversed phase-encoding for distortion correction)
- Gradient-echo magnetization transfer (MT) 3D (3:34 min; TR = 28 ms, TE = 3.3 ms, flip angle = 6°, 1.5 mm isotropic)
- Gradient-echo proton density (PD) 3D (same parameters as MT, without the MT pulse)
- Gradient-echo T1-weighted 3D (same parameters as MT, except TR = 18 ms, flip angle = 20°)
- MP2RAGE 3D (7:26 min; TR = 4 s, TE = 1.51 ms, TI1 = 700 ms, TI2 = 1500 ms, flip angle 1 = 7°, flip angle 2 = 5°, 1.2 mm isotropic, R = 2)
- Susceptibility-weighted 3D (4:54 min; TR = 27 ms, TE = 20 ms, flip angle = 15°)

Spinal cord anatomical MRI

Spinal cord sessions followed the community spine-generic standard protocol (Cohen-Adad et al., 2021) and were acquired during the same dedicated anatomical sessions as brain data. The protocol included T1-weighted (3D sagittal, 1.0 mm isotropic), T2-weighted (3D sagittal, 0.8 mm isotropic), diffusion-weighted (2D axial, cardiac-gated, $0.9 \times 0.9 \times 0.5$ mm), magnetization transfer, proton density, T1-weighted, and multi-echo gradient-echo (3D axial, $0.9 \times 0.9 \times 0.5$ mm) sequences. Full acquisition parameters are documented in the BIDS dataset metadata.

Stimulus Delivery

Visual stimuli were projected onto a screen in the MRI room using an Epson Powerlite L615U projector via a waveguide. For most datasets, audio was delivered via S15 MRI-compatible earphone inserts (Sensimetrics), with a custom impulse response applied online using a finite impulse response filter to correct for headphone frequency response. Sound was amplified using an AudioSource AMP100V amplifier located in the control room.

Two successive hearing protection configurations were used across the project. The initial setup combined S15 earphone inserts, disposable Comply canal tips (NRR 29 dB), and modified commercial earmuffs (NRR 27 dB; Stanley) trimmed to fit inside the head coil. This setup was used for the `hcptrt`, `movie10`, `friends` seasons 1–4, and `shinobi` datasets. It was subsequently replaced due to pressure discomfort during extended sessions. The current setup uses S15 earphone inserts with custom-sized Comply canal tips selected per participant, combined with memory foam headphone rings (Brainwavz Audio), and was used from `friends` season 5 onward.

Task stimuli were presented using a custom overlay built on top of PsychoPy (Peirce et al., 2019) for all datasets except `hcptrt`, for which E-Prime scripts adapted from the Human Connectome Project were used. Stimulus scripts are publicly available at https://github.com/courtois-neuromod/task_stimuli.

Physiological Recordings

During all scanning sessions, physiological signals were recorded using a Biopac M160 MRI-compatible system at 1000 Hz, synchronized to the MRI scanner via trigger pulses and monitored with AcqKnowledge software. The following signals were recorded:

- **Cardiac (plethysmograph):** A Biopac TSD200-MRI photoplethysmogram transducer placed on the foot or toe provided beat-by-beat heart rate estimates.
- **Electrocardiogram:** Three MRI-compatible electrodes placed on the lower left rib cage recorded cardiac electrical activity.
- **Skin conductance:** Two electrodes applied to the sole and ankle of the foot recorded electrodermal activity.
- **Respiration:** A custom MRI-compatible respiration belt consisting of a blood pressure cuff, a pressure sensor (MPXV5004GC7U, NXP), and flexible tubing measured thoracic pressure via an analog input to the Biopac system.

In later datasets (see [DATA RECORD TABLE]), eye movements and pupil dilation were recorded using a high-speed infrared camera (MRC Systems) with illuminating IR LEDs; gaze data were processed with Pupil open-source software.

Data Record

All CNeuroMod data are distributed as DataLad repositories hosted on GitHub under the courtois-neuromod organisation. The master meta-repository, `cneuromod.all`, tracks all datasets and their derivatives as git submodules following YODA principles. Full technical documentation is available at docs.cneuromod.ca. Data are released under a CC0 license.

Repository structure

`cneuromod.all` is a DataLad YODA meta-repository comprising 43 git submodules. Each experimental paradigm occupies a top-level folder (e.g., `friends/`, `hcptrt/`, `anat/`), which in turn contains independent submodules for each data component:

Submodule suffix	Contents
`<dataset>/bids`	Raw BIDS data
`<dataset>/fmriprep`	fMRIPrep preprocessing derivatives
`<dataset>/mriqc`	MRIQC quality-control reports
`<dataset>/physprep`	Physiological preprocessing derivatives
`<dataset>/<other>`	Dataset-specific additional content

Most data files are git-annex symlinks and must be explicitly retrieved with `datalad get`. Cloning the meta-repository fetches only metadata (git history, file pointers); data files are downloaded on demand. Recursive installation of submodules should be avoided, as submodules re-expose their own sub-submodules at differing versions for provenance tracking.

BIDS compliance

All functional and anatomical neuroimaging data are formatted according to the Brain Imaging Data Structure (BIDS) specification. Deviations from the core specification are:

- Multi-contrast anatomical sequences follow BEP001.
- Spinal cord images use the Body Part tag proposed in BEP025 (`bp-cspine`) to distinguish them from brain anatomical images acquired with the same contrasts.

Session indices (`ses-001`, `ses-002`, ...) reflect the order of data acquisition; the number of runs, tasks, and their order within a session vary across participants. A small number of session indices are skipped where an entire session was discarded for scanning issues.

Anatomical data (`anat/`)

The anatomical dataset contains longitudinal brain and spinal cord MRI collected at approximately four sessions per year across the full duration of the project. All images covering the face were anonymised by zeroing the face, teeth, and ear regions with a custom mask warped from MNI space.

Submodules: `anat/bids` (raw), `anat/smriprep`, `anat/smriprep.longitudinal`, `anat/freesurfer`, `anat/freesurfer.longitudinal`, `anat/atlases`, `anat/pycortex`.

Functional datasets

friends (~65 h/subject, N = 6)

Participants watched all seven seasons of the American sitcom *Friends* in English. Each episode is split into two scanning segments (a/b). The BIDS `task` entity encodes season, episode, and segment (e.g., `task-s01e01a`). Season 7 fMRI responses are withheld as a held-out test set for the Algonauts Project 2025 Challenge. Physiological recordings (ECG, respiration, plethysmograph, EDA) accompany all runs. **Submodules:** `friends/bids`, `friends/fmriprep`, `friends/mriqc`, `friends/physprep`.

hcptrt (~9 h/subject, N = 6)

Participants completed 13–18 repetitions of the seven Human Connectome Project (HCP) task-fMRI localizers, yielding 36 standard GLM contrasts, plus 4–6 resting-state runs per subject. Stimuli and E-Prime scripts are adapted from the HCP and available at the HCP database. **Submodules:** `hcptrt/bids`, `hcptrt/fmriprep`.

movie10 (~10 h/subject, N = 6)

Participants watched four feature films (*The Bourne Supremacy*, *The Wolf of Wall Street*, *Life*, and *Hidden Figures*), each cut into approximately 10-minute segments. Physiological recordings accompany all runs. **Submodules:** `movie10/bids`, `movie10/fmriprep`, `movie10/mriqc`, `movie10/physprep`.

mario (~17 h/subject, N = 5)

Participants played *Super Mario Bros.* (Nintendo, 1985) across 22 levels in a structured discovery phase followed by a randomised practice phase. Physiological recordings accompany all runs. **Submodules:** `mario/bids`, `mario/fmriprep`, `mario/physprep`.

shinobi (~8 h/subject, N = 4)

Participants played *Shinobi III: Return of the Ninja Master* (Sega, 1993) in two phases. Physiological recordings accompany all runs. Training-session behavioural data are available in `shinobi/training`. **Submodules:** `shinobi/bids`, `shinobi/fmriprep`, `shinobi/mriqc`, `shinobi/physprep`, `shinobi/training`.

things (~16 h/subject, N = 4)

Participants completed a continuous visual recognition task with images drawn from the THINGS object database (~4320 unique images per participant from 720 categories; each image presented 3× across sessions). GLM and GLMsingle first-level estimates are provided as derivatives. **Submodules:** `things/bids`, `things/fmriprep`, `things/mriqc`, `things/behaviour`, `things/glm`, `things/glmsingle`.

harrypotter (~1.2 h/subject, N = 5)

Participants read chapter 9 of *Harry Potter and the Sorcerer's Stone* word-by-word at 2 Hz across 7 runs. Stimuli are adapted from (**wehbe2014**). Physiological recordings accompany all runs. **Submodules:** `harrypotter/bids`, `harrypotter/fmriprep`, `harrypotter/physprep`.

floc (~0.8 h/subject, N = 4)

A functional localizer task (adapted from the Stigliani, Weiner, and Grill-Spector (2015)) identifying brain regions selective for five visual categories (faces, bodies, places, objects, characters) across 6 sessions × 2 runs. Nine GLM contrasts and subject-specific ROIs (FFA, OFA, pSTS, PPA, OPA, MPA, EBA) are provided. **Submodules:** `floc/bids`, `floc/fmriprep`, `floc/mriqc`, `floc/rois`.

retinotopy (~1.4 h/subject, N = 4)

A retinotopy task (adapted from (**kay2013**)) using ring, bar, and wedge apertures across 5–6 sessions × 3 runs to derive population receptive field (pRF) maps and delineate 12 visual ROIs (V1–V3, hV4, VO1/2, LO1/2, TO1/2, V3a/b). **Submodules:** `retinotopy/bids`, `retinotopy/fmriprep`, `retinotopy/prf`.

gamepad (~0.7 h/subject, N = 4)

A cued-response button-press task validating the CNeuroMod MRI-compatible fiber-optic gamepad (**harel2023**). **Submodule:** `gamepad/bids`, `gamepad/fmriprep`.

Preprocessing derivatives

fMRIPrep

Functional data were preprocessed using fMRIPrep 20.2.5 (**esteban2019**) via an anatomical fast-track using sMRIPrep output (`--anat-derivatives`), ensuring a consistent anatomical basis across all functional datasets. The `--ignore slicetiming` flag was used. Outputs are provided in three spaces: native T1w, volumetric MNI152NLin2009cAsym, and surface-based fsLR-den-91k (grayordinates). Each functional run yields:

- `*_desc-preproc_bold.nii.gz` — preprocessed BOLD timeseries
- `*_boldref.nii.gz` — single-volume BOLD reference
- `*_desc-brain_mask.nii.gz` — brain mask
- `*_desc-confounds_timeseries.tsv` — confound regressors (motion parameters, CompCor components, global signals, framewise displacement, DVARS)

sMRIPrep / FreeSurfer

Anatomical processing used sMRIPrep and FreeSurfer in both cross-sectional and longitudinal modes, providing cortical surface reconstructions, segmentations, and subject-specific atlases. Cortical flat maps are available via PyCortex surfaces (`anat/pycortex`).

PhysPrep

Physiological signals were preprocessed using the CNeuroMod PhysPrep pipeline (**PhysPrep**), integrating Phys2Bids, NeuroKit2, and Systole. Outputs per run include:

- `*_physio.tsv.gz` / `*_physio.json` — raw segmented biosignals (in `<dataset>/bids`)
- `*_desc-preproc_physio.tsv.gz` — filtered and cleaned timeseries
- `*_desc-physio_events.tsv` — sparse extracted features (peaks, troughs, SCRs)
- `*_desc-quality.json` — run-level quality assessment (pass/fail per modality)

Data access

Five of the six participants (`sub-01`, `sub-02`, `sub-03`, `sub-05`, `sub-06`) have consented to fully open sharing via the Canadian Open Neuroscience Platform (CONP). Access to all six participants requires a registered-access data transfer agreement (DTA), obtainable via cneuromod.ca/access. Approved researchers receive S3 credentials for download. Full download instructions are available at docs.cneuromod.ca.

Versioning

`cneuromod.all` uses yearly release tags (e.g., `git checkout 2020`) to pin a specific state of all submodule pointers, enabling exact reproduction of prior analyses. Updates to an existing clone can be obtained with `datalad update -r --merge --reobtain-data`.

Technical Validation

[Demonstrate the quality and reliability of the data through quantitative quality metrics, reproducibility analyses, and comparisons against established benchmarks.]

fMRI data quality

Functional image quality was assessed with MRIQC (Esteban et al., 2017), computed per run on the raw BOLD data. We report head-motion and temporal signal-to-noise ratio (tSNR) image-quality metrics (IQMs) across the 4,537 functional runs with released MRIQC derivatives, spanning 17 datasets, all six participants, and 880 unique dataset × subject × session combinations. Four datasets (`anat`, `emotion-videos`, `langlocalizer`, `mario`) have no BOLD MRIQC derivatives available yet and are excluded from this analysis; the numbers below therefore describe the datasets with released MRIQC derivatives, not the full collection. No pass/fail threshold is applied anywhere in this analysis and no run is excluded — the distributions below describe the released data as-is.

Head motion, summarized as mean framewise displacement (FD) per run (Power et al., 2012), was low overall: median 0.145 mm (mean 0.147, SD 0.062, range 0.046–0.480 mm across runs). No run exceeded a mean FD of 0.5 mm, and 787 runs (17.3%) exceeded 0.2 mm. At the volume level (3,582 runs with MRIQC framewise-displacement timeseries available), a median of 21.7% of volumes per run exceeded 0.2 mm and a median of 0.5% exceeded 0.5 mm (means 23.2% and 2.1%). Motion varied about 2.5-fold across participants — sub-03 and sub-04 showed the lowest motion (median FD 0.074 and 0.081 mm), sub-02 and sub-06 the highest (0.178 and 0.187 mm) — while sub-01 and sub-05 fell in between (0.122 and 0.132 mm). Motion also varied systematically with task demands: passive paradigms such as `floc`, `movie10`, and `hcptrt` had the lowest median FD (0.111–0.118 mm), while the video-game datasets `mario3`, `shinobi`, and `mariostars` had the highest (0.177–0.206 mm), consistent with the additional head movement associated with active gameplay.

Temporal SNR followed the inverse pattern: median 29.8 across runs (mean 29.6, SD 5.3, range 11.5–40.1), with a strong negative correlation between per-run mean FD and tSNR (Pearson r = -0.76, Spearman ρ = -0.77) — runs and participants with more motion have correspondingly lower tSNR. Per-subject median tSNR ranged from 26.5 (sub-06) to 35.2 (sub-03), and per-dataset median tSNR from 22.3 (`shinobi`) to 33.0 (`hcptrt`).

Regional tSNR was further characterized by averaging per-run tSNR maps within a combined Schaefer-1000/7-network cortical (Schaefer et al., 2018; Yeo et al., 2011), Tian-S3 subcortical (Tian et al., 2020), and Nettekoven cerebellar (Nettekoven et al., 2024) atlas, collapsed to 11 region groups. This analysis covers 936 runs from the `floc`, `retinotopy`, and `things` datasets only, as the per-run tSNR maps for the remaining datasets sit on credentialed data remotes that were not retrievable for this pass; the regional pattern below should be read as illustrative of the acquisition's spatial signal-quality profile rather than as a comprehensive summary. Median tSNR was lowest in the Limbic network (18.7 — orbitofrontal and ventral-temporal cortex) and in subcortical structures (thalamus 27.4, caudate 28.6, putamen 29.0) and cerebellum (30.7), and highest in dorsal cortical networks (Dorsal Attention 46.5, Control 43.6, Somatomotor 43.4, Salience/Ventral Attention 42.6, Default 40.7, Visual 37.7). This ordering reflects the expected susceptibility-dropout pattern of gradient-echo EPI near air-tissue interfaces, a property of the acquisition geometry rather than of CNeuroMod specifically; users of ventral-temporal, orbitofrontal, or deep subcortical signal should budget for reduced tSNR in these regions.

Taken together, these metrics indicate low and stable head motion and adequate temporal signal quality across the released functional runs, with expected, interpretable variation across participants, tasks, and brain regions. The main limitations of this first-pass analysis are its restriction to datasets with released MRIQC and tSNR derivatives, and the regional breakdown's reliance on a three-dataset subset; both will be extended as further derivatives become publicly available.

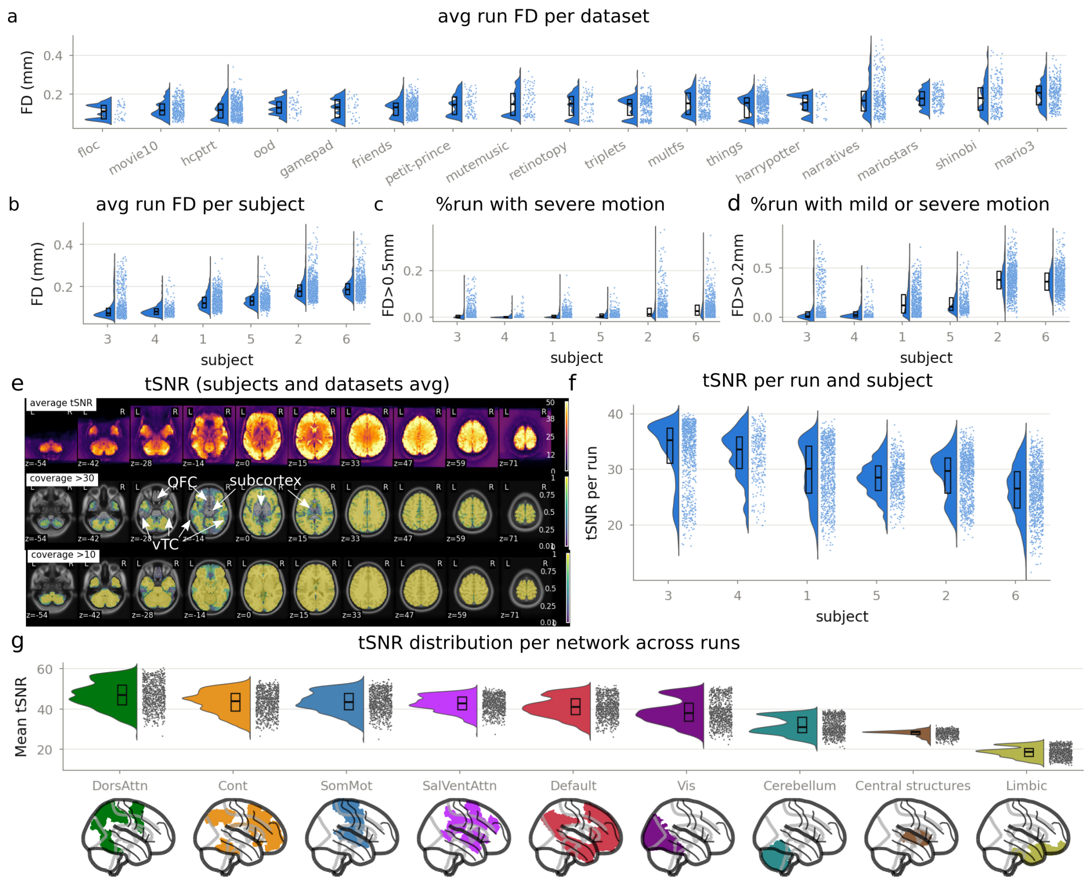

Figure 2. fMRI data quality across the CNeuroMod datasets. (a) Average run FD per dataset. **(b)** Average run FD per subject. **(c)** Percentage of runs with severe motion (mean FD >0.5 mm) per subject. **(d)** Percentage of runs with mild or severe motion (mean FD >0.2 mm) per subject. **(e)** Average tSNR maps across subjects and datasets (top), with voxelwise coverage maps thresholded at tSNR >30 (middle) and tSNR >10 (bottom); orbitofrontal cortex (OFC), ventral temporal cortex (vTC), and subcortex are annotated as regions of reduced coverage. **(f)** tSNR per run, by subject. **(g)** tSNR distribution per region group across runs (`floc`, `retinotopy`, `things`), from worst (Limbic) to best (Dorsal Attention), with matching glass-brain maps of each region group below.

sMRI data quality

Structural image quality was assessed separately across the anatomical acquisitions of the same six participants and is reported in a companion publication [MISSING REF: companion CNeuroMod structural data quality paper — citation to be supplied].

Reproducibility

[Show test-retest reliability or within-subject consistency across sessions.]

Preprocessing Pipeline Validation

[Describe any validation steps applied to preprocessed derivatives.]

Usage Notes

The CNeuroMod dataset has been used in a growing body of research spanning brain encoding, brain decoding, cognitive neuroscience, and AI alignment with neural data. This section provides an overview of key research directions enabled by the dataset and practical guidance for new users.

Overview of Publications

[Summary paragraph: briefly enumerate the main research communities using CNeuroMod data, total number of publications or preprints to date, and the main venues (NeurIPS, ICLR, Imaging Neuroscience, PLOS ONE, etc.).]

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1. Individual Brain Encoding Models

[Overview: CNeuroMod has emerged as a key resource for building high-quality voxelwise encoding models of individual brains.]

The Algonauts 2025 Competition

[Describe the Algonauts 2025 challenge: goals, dataset used (CNeuroMod HCP-style or Friends/Shinobi?), number of participants, main findings. Highlight how the competition structure promoted individual-level modeling.]

The TRIBE Model

[Describe the TRIBE model: what it does, what data it was trained on, key results in terms of brain prediction accuracy. Reference the relevant paper(s).]

New Directions: Auto-Regressive and Brain Encoding in One Model

[Discuss Paugam et al.'s work combining auto-regressive modeling with brain encoding in a unified framework. Explain why this is a significant conceptual advance and what it implies for future encoding model architectures. Add citation.]

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2. Brain Encoding Models of the Active Brain

[Overview: CNeuroMod's active tasks — particularly videogame paradigms — enable a new class of encoding models that capture brain activity during goal-directed, embodied behavior.]

Atari Games

[Discuss Cross et al. and Tomov et al. using Atari game stimuli for brain encoding. Summarize key findings linking RL agent representations to neural activity. Add citations.]

Videogame Controller and Motor Signals

[Discuss Harel et al. (PLOS ONE) examining controller inputs and neural correlates. Summarize findings on motor and planning signals in fMRI. Add citation.]

Clean BOLD Signal in Active Tasks

[Describe Harel et al. (Imaging Neuroscience) demonstrating that high-quality BOLD signal is recoverable during active gameplay despite motion and arousal confounds. Add citation.]

Imitation Learning in the Brain

[Describe Kemtur et al. (Imaging Neuroscience) using imitation learning frameworks to model behavior and brain activity. Add citation.]

Learning Trajectories in Mario

[Describe Harel & Bellec (RLC Workshop on Videogames in RL) characterizing how neural representations evolve as subjects learn to play Super Mario Bros. Add citation. Note this as a major area for future competitions.]

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3. Towards Better AI with Neural Data

[Overview: Growing excitement in the ML community about using neural recordings as a source of inductive biases or alignment signals for AI models.]

The Case for Neuro-Aligned AI

[Reference Mineault et al. white paper and recent review articles arguing for integrating neural data into AI training pipelines. Summarize the main arguments.]

SoundNet Trained on Neural Data

[Describe Freteault et al. (Imaging Neuroscience) training a SoundNet-style audio model using CNeuroMod fMRI responses as supervision signal. Summarize results showing improved audio representations. Add citation.]

Challenges and Proper Downstream Evaluation

[Discuss the challenge of limited neural data relative to large model parameter counts. Argue that benchmarking on fine-tuning on small datasets is more meaningful than competing on large-scale benchmarks with unconstrained compute. Provide practical recommendations for future neuro-AI work using CNeuroMod.]

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4. Brain Decoding

[Overview: CNeuroMod supports brain decoding — reconstructing stimuli or mental states from neural activity — across multiple modalities and task types.]

Decoding with the Shinobi Dataset

[Describe Shima et al. (Imaging Neuroscience) using the Shinobi videogame fMRI dataset for brain decoding. Summarize what was decoded (game state? actions? rewards?) and key findings. Add citation.]

Expanding the Space of Brain Decoding

[Explain how the breadth of CNeuroMod stimuli and tasks greatly expands the stimulus and cognitive spaces available for decoding, beyond the traditional image/language domains.]

Upcoming Benchmarks

[Preview the triplets dataset (for word-level semantic decoding) and the MultiFS working memory dataset as upcoming resources that will open new decoding benchmarks for the community.]

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5. Cognitive Neuroscience and Naturalistic Annotations

[Overview: CNeuroMod naturalistic stimuli, combined with rich semantic annotations, support traditional cognitive neuroscience questions about emotion, memory, language, and social cognition.]

Emotion Annotations in Friends

[Highlight recent works using emotional annotations synchronized with the Friends TV show fMRI data. Summarize the types of annotations available and key cognitive neuroscience findings. Add citations.]

Large-Scale Annotation Efforts

[Describe the team's ongoing effort to release large-scale annotations of naturalistic stimuli, including Friends and Mario scenes (scene segmentation, character identity, emotional valence, actions, etc.). Explain how these annotations will enable purely cognitive neuroscience-driven analyses without requiring ML expertise.]

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6. Foundation Models and the Digital Brain

[Overview: CNeuroMod individual brain models are positioning themselves as building blocks for foundation models that generalize across individuals, datasets, acquisition sites, and modalities.]

TRIBE V2 and the Digital Brain Project

[Describe TRIBE V2 and its role in the broader Digital Brain initiative. Explain how scaling individual encoding models across CNeuroMod subjects and linking to other large datasets (e.g., HCP, UK Biobank) could yield a generalizable neural foundation model. Add citations/links.]

Optimal Transport for Data-Efficient Alignment

[Discuss the opportunity to apply optimal transport (OT) methods for aligning neural representations across subjects and datasets in a data-efficient manner. Reference work from Aimy Wenegrat's lab and the INRIA DANDI team. Explain why OT is particularly well-suited to the small-N, high-dimensional regime of deep phenotyping datasets like CNeuroMod.]

Accessing the Data

[Instructions for requesting access and downloading data via the CNeuroMod data portal. Include links to the data agreement, DataLad/datalad-cneuromod repository, and OpenNeuro/OSF deposits where applicable.]

Known Limitations

[Describe known issues: small N (6 subjects), site-specific scanner characteristics, missing sessions for some subjects/tasks, motion in active paradigms, and task-specific exclusion criteria. Refer readers to the Data Record section for per-dataset details.]

Recommended Practices

[Suggest best practices: use fMRIPrep outputs, leverage provided confound regressors, cite the relevant task papers when using specific datasets, check the CNeuroMod documentation for versioned data releases.]

Data Availability

[State where the data are deposited, under what license, and how to access them. Include DOIs and links to data portals.]

Code Availability

[List all software repositories associated with data collection, preprocessing, and quality control. Include version numbers and DOIs where available.]

The depth-vs-breadth dataset comparison figure (Figure 1) was generated with `courtois-neuromod/datase`

The fMRI data quality analysis and figure (Figure 2) were generated with `courtois-neuromod/cneuromod.al`

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Author Contributions

[List each author's specific contribution using CRediT taxonomy roles: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.]

Competing Interests

[Declare any competing interests, or state that the authors declare no competing interests.]

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